

Nexus Enterprise Digital Transformation Programme

Master Programme Charter ? Version 3.0

1. Executive Summary

The Nexus Programme is the largest technology transformation initiative ever undertaken by the organisation.

It replaces five legacy systems with a unified cloud-native architecture spanning Identity, Data, Network, AI, and Compliance domains.

The programme is structured across four sequential phases: Foundation (February 2026 to June 2026), Integration (May 2026 to September 2026), Validation (July 2026 to November 2026), and Production Deployment (October 2026 to March 2027).

Total programme budget is £47.2 million across 22 months.

The programme is governed by the Nexus Steering Board, reporting to the Chief Technology Officer.

All phase gates require sign-off from the Compliance Authority and the Security Certification Body before any production-affecting activity may proceed.

This charter is effective from 1 February 2026 and supersedes all prior programme documentation.

2. Programme Objectives and Strategic Alignment

The Nexus Programme delivers against four strategic pillars endorsed by the Board of Directors in the 2025 Technology Strategy Review.

Pillar 1 · Unified Identity: Every internal system and customer-facing service must authenticate through a single Identity Platform, replacing seven disparate login mechanisms and eliminating the password synchronisation failures that caused three production incidents in 2025.

Pillar 2 · Data Consolidation: All transactional and analytical workloads must be served from a single Core Database cluster, eliminating the twelve point-to-point data feeds currently maintained by the Database Administration Team.

Pillar 3 · Infrastructure Modernisation: The ageing on-premises network must be replaced by a fully resilient Network Infrastructure Upgrade, enabling Cloud Migration and the retirement of the primary data centre lease which expires on 31 December 2026.

Pillar 4 · Intelligent Automation: An AI Platform fed by a governed Training Data Repository must be operational by Q4 2026, enabling the Machine Learning Pipeline

that the Product organisation has committed to three enterprise customers in binding SLAs.

3. Programme Governance Structure

The Nexus Steering Board has ultimate authority over scope, budget, and schedule. It meets fortnightly and produces the Programme Authority Notice consumed by the Compliance Authority at each phase gate.

The Compliance Authority requires the Programme Authority Notice before opening any phase gate review.

The Compliance Authority produces the Gate Approval Certificate consumed by every delivery workstream as the prerequisite for production activity.

The Security Certification Body operates independently of the Nexus Steering Board and requires a completed penetration test report before issuing the Security Certification.

The Security Certification is required by Cloud Migration, by Production Deployment, and by the AI Platform go-live.

The Data Protection Officer produces the GDPR Compliance Sign-Off required by the Training Data Repository before any personal data may be ingested for model training.

The Change Advisory Board requires a completed impact assessment from each workstream before scheduling any change in the production change window.

Programme-level risks are owned by the Programme Director, who reports to the Nexus Steering Board at each fortnightly session.

4. Identity Domain ? Workstream A

The Identity Platform is the central authentication and authorisation broker for all Nexus programme workstreams.

It provides OAuth 2.0, SAML 2.0, and OpenID Connect endpoints consumed by every downstream application.

The SSO Module requires Identity Platform to be deployed and health-checked before any user-facing application may be connected.

The MFA Gateway requires Identity Platform to be operational because all second-factor challenges are routed through the Identity Platform token service.

The HR Integration Service requires Identity Platform so that employee provisioning and deprovisioning events are propagated to all connected systems within the SLA of four hours.

The Finance Integration Module requires Identity Platform for role-based access control enforcement on all financial transactions above £10,000.

The Customer Portal requires Identity Platform because unauthenticated access to customer account data is prohibited under the organisation's data classification policy.

The Audit System requires Identity Platform to capture a complete authentication event log for the mandatory 7-year retention period required by the Financial Conduct Authority.

Identity Platform is therefore the highest-centrality node in the programme dependency graph: six separate workstreams cannot proceed until it is live.

The Identity Platform is scheduled to be deployed and signed off by 1 April 2026.

5. Identity Domain ? Open Conflict: Privacy Board Objection

The Privacy Board has formally objected to the Identity Platform architecture on the grounds that centralising all authentication events in a single platform creates a high-value target for data exfiltration and violates the data minimisation principle of the UK GDPR.

The Privacy Board blocks Identity Platform deployment until a privacy impact assessment is reviewed and approved, a process estimated to take between 60 and 90 days from submission.

The SSO Module requires Identity Platform, and the Identity Platform deployment is blocked by the Privacy Board.

The MFA Gateway requires Identity Platform, which is blocked.

The HR Integration Service requires Identity Platform, which the Privacy Board blocks.

The Finance Integration Module requires Identity Platform; the Privacy Board blocks it.

The Customer Portal requires Identity Platform, which remains blocked.

This constitutes a cascading structural conflict: every workstream that requires Identity Platform is transitively blocked by the Privacy Board's objection.

The Nexus Steering Board is aware of this conflict.

As of the date of this charter, no resolution or escalation path has been agreed.

The programme schedule assumes Identity Platform deployment proceeds on 1 April 2026, which is inconsistent with the Privacy Board's stated timeline.

Resolution options include federated identity (rejected by Security Certification Body as insufficiently auditable) or a phased privacy review (not yet scheduled).

6. Data Domain ? Workstream B

The Core Database cluster is a PostgreSQL 16 active-active deployment across three availability zones, replacing Oracle 11g instances running on hardware that reaches end-of-support on 30 June 2026.

The Analytics Engine requires Core Database because all analytical queries are issued directly against the production replica set; a separate analytical database was rejected as a cost overrun in the 2025 budget review.

The Reporting Module requires Core Database for its nightly batch extract jobs that populate the executive dashboards consumed by the Nexus Steering Board.

The Customer Portal requires Core Database to retrieve and persist customer account records in real time.

The API Gateway requires Core Database for request routing rules and rate-limit configuration stored in the operations schema.

The Archive System requires Core Database because the archival policy engine reads retention metadata from the Core Database catalogue table.

Data migration from Oracle 11g to PostgreSQL 16 is scheduled to begin on 1 April 2026 and complete on 31 July 2026.

Core Database cluster provisioning begins on 1 February 2026 and must be complete by

31 March 2026 so that migration tooling can be validated before the migration window opens.

7. Data Domain ? Open Conflict: Data Centre Decommission

Cloud Migration requires Data Centre Decommission to proceed on schedule, because the cloud hosting costs and the data centre lease cannot be carried concurrently beyond 31 December 2026 without breaching the programme budget envelope.

Data Centre Decommission requires all production workloads to have been migrated to cloud infrastructure and validated before the facility access is surrendered.

However, the Compliance Authority blocks Data Centre Decommission because the on-premises audit trail · stored on data centre-local storage arrays · must remain accessible for a minimum of 12 months following the end of the last financial year covered by the audit.

The Compliance Authority's position is that decommissioning the data centre before 31 March 2027 would breach statutory record-keeping obligations.

Cloud Migration requires Data Centre Decommission, but the Compliance Authority blocks Data Centre Decommission.

This is a direct structural conflict.

The data centre lease expiry on 31 December 2026 creates a hard cost forcing function that the Compliance Authority objection does not accommodate.

The Legal team has been asked to advise on whether cloud-replicated audit logs satisfy the statutory requirement.

No response has been received as of the charter date.

8. Network Domain ? Workstream C

The Network Infrastructure Upgrade replaces the organisation's flat 10GbE LAN with a fully segmented 100GbE spine-and-leaf topology with software-defined networking control.

The upgrade is scheduled to begin on 1 March 2026 and complete by 31 May 2026.

Cloud Migration requires Network Infrastructure Upgrade because the direct-connect circuits to the cloud provider require the new spine switches to be in place.

Video Conferencing Platform requires Network Infrastructure Upgrade because the existing network cannot sustain the QoS policies required for 4K video across 40 concurrent rooms.

Secure Remote Access requires Network Infrastructure Upgrade because the zero-trust NAC policy enforcement points are integrated into the new spine-and-leaf control plane.

VPN Gateway requires Network Infrastructure Upgrade because the IPsec tunnel termination capacity is physically hosted on the new spine switches.

The Network Infrastructure Upgrade is therefore a hub node: four separate workstreams are blocked until it completes.

Any delay to the network upgrade propagates to Cloud Migration, Video Conferencing, Secure Remote Access, and VPN Gateway simultaneously.

The network team has flagged a 15% risk of a 4-week delay due to supplier lead times on the 100GbE line cards, which are currently on a 14-week order cycle.

9. Network Domain ? Open Conflict: Security Certification Hold

The Security Certification Body requires a completed penetration test of the new network architecture before issuing the Security Certification that permits Cloud Migration.

The penetration test requires the Network Infrastructure Upgrade to be fully installed and operational before the test environment can be stood up.

Cloud Migration requires Security Certification, and Security Certification requires the penetration test, and the penetration test requires Network Infrastructure Upgrade.

However, the Security Certification Body blocks Cloud Migration until certification is issued, and the Security Certification Body has stated it will not issue certification based on a partial or simulated network environment.

The programme schedule assumes Cloud Migration begins on 1 June 2026.

The Security Certification Body's earliest available penetration test slot is 15 June 2026, after the Network Infrastructure Upgrade completes on 31 May 2026.

The penetration test and remediation cycle is estimated at 6 to 8 weeks, meaning Security Certification cannot be issued before 10 August 2026 at the earliest.

Cloud Migration requires Security Certification; the Security Certification Body blocks Cloud Migration until certification is issued.

The scheduled Cloud Migration start date of 1 June 2026 is therefore unachievable given the Security Certification Body's constraints.

This conflict has not been escalated to the Nexus Steering Board.

10. AI Domain ? Workstream D

The AI Platform is a GPU-accelerated model training and inference cluster deployed on cloud infrastructure.

It exposes a managed API consumed by the Machine Learning Pipeline and the BI Reporting Suite's predictive analytics module.

The AI Platform requires the Training Data Repository to be populated with at least 18 months of historical transaction data before the first model training run can produce results within the accuracy thresholds committed to enterprise customers.

The Machine Learning Pipeline requires AI Platform because all feature engineering, model training, and inference serving is delegated to the platform's managed runtime.

The BI Reporting Suite requires the Data Warehouse, and the Data Warehouse requires Core Database as its source-of-record feed.

The AI Platform is scheduled to be operational by 1 August 2026.

The Training Data Repository population is scheduled to begin on 1 May 2026 and complete by 31 July 2026.

The Machine Learning Pipeline first production run is scheduled for 1 September 2026.

Three enterprise customers have contracted for ML Pipeline outputs from 1 October 2026, creating a hard external deadline that cannot be moved without triggering penalty clauses totalling £3.2 million.

11. AI Domain ? Open Conflict: GDPR Training Data Block

The Training Data Repository requires GDPR Compliance Sign-Off from the Data Protection Officer before any personal data may be ingested.

The repository contains customer transaction records, browsing behaviour, and support interaction logs - all of which are classified as personal data under the UK GDPR definition.

The GDPR Compliance Framework blocks Training Data Repository population until a lawful basis for processing has been established, a Records of Processing Activities entry has been created, and a Data Protection Impact Assessment has been signed off by the Data Protection Officer.

The Data Protection Officer has estimated the DPIA review and sign-off process will take a minimum of 10 weeks from the date of submission.

The DPIA has not yet been submitted as of the date of this charter.

The AI Platform requires Training Data Repository.

The GDPR Compliance Framework blocks Training Data Repository.

Therefore the AI Platform cannot meet its 1 August 2026 go-live date unless the DPIA is submitted immediately and the Data Protection Officer prioritises its review.

The Machine Learning Pipeline requires AI Platform, meaning the 1 September 2026 first production run is also at risk.

The enterprise customer penalty exposure of £3.2 million is directly contingent on resolving this GDPR compliance conflict before 1 July 2026.

12. Infrastructure Domain ? Workstream E

The infrastructure workstream decommissions the legacy on-premises estate and migrates all workloads to the contracted cloud provider.

The Disaster Recovery System requires the Primary Production System to be the source of replication; DR cannot be stood up until the Primary Production System has been migrated to cloud and health-checked.

Disaster Recovery System provisioning is scheduled to begin on 1 July 2026 and complete by 31 August 2026.

Primary Production System cloud migration is scheduled to begin on 1 June 2026 and complete by 30 June 2026.

The Container Orchestration platform requires the DevOps Pipeline to be operational because all containerised workload deployments are managed through the CI/CD toolchain.

The DevOps Pipeline requires the API Gateway to be deployed before pipeline jobs can push artefacts to the container registry.

The Service Mesh requires Container Orchestration because all inter-service mTLS is managed by the service mesh control plane running inside the cluster.

Load Balancer Cluster requires API Gateway to be deployed and traffic routing rules to be configured before any external traffic can be admitted to the platform.

The infrastructure workstream is on the critical path for 14 of the 22 delivery milestones in the programme schedule.

13. Infrastructure Domain ? Open Conflict: SLA Uptime Block

The Primary Production System migration to cloud requires a maintenance window during which the primary database is in read-only mode for an estimated 4 to 6 hours.

The SLA Compliance obligation requires 99.95% monthly uptime for the Primary Production System, which permits a maximum of 21.6 minutes of unplanned downtime per month.

The SLA Compliance obligation blocks Primary Production System Shutdown because a 4-to-6-hour maintenance window would consume 1,100% of the monthly downtime allowance in a single event, triggering SLA breach penalties estimated at £180,000 per hour.

Cloud Migration requires the Primary Production System to be migrated, which requires a shutdown window, which the SLA Compliance obligation blocks.

Disaster Recovery System requires Primary Production System to be on cloud infrastructure, meaning the DR go-live is also transitively blocked.

The Infrastructure team has proposed a blue-green migration approach that would reduce the downtime window to under 5 minutes, but this approach requires a second cloud environment licence costing £240,000 that has not been approved in the programme budget.

The Nexus Steering Board has not been presented with this trade-off.

14. API and Integration Domain ? Workstream F

The Unified API Layer consolidates 23 legacy point-to-point integrations into a single managed API gateway, reducing integration maintenance effort from 3.4 FTE to 0.8 FTE annually.

The Unified API Layer requires Legacy API Deprecation to be completed because the legacy endpoints and the new Unified API Layer cannot coexist on the same hostname without routing conflicts that would make end-to-end tracing impossible.

The API Gateway requires Core Database for its configuration store, as described in Workstream B.

The Third-Party Integration Framework connects 14 external vendors to the organisation's APIs under contractual arrangements that reference specific legacy endpoint URLs.

The Third-Party Integration Framework blocks Legacy API Deprecation because vendor contracts contain clause 7.4 which requires 12 months' written notice before any endpoint URL change.

The earliest any vendor contract permits endpoint migration is 1 March 2027.

The Unified API Layer requires Legacy API Deprecation, and the Third-Party Integration Framework blocks Legacy API Deprecation.

This is a direct structural conflict.

The programme schedule assumes Legacy API Deprecation completes by 30 September 2026, which is 18 months before the contractual notice period permits it.

Legal has been asked to review clause 7.4 but has not yet provided an opinion.

15. Change Management and Communications

The Change Management workstream supports all five delivery workstreams with stakeholder communications, training delivery, and adoption measurement.

The HR Integration Service deployment requires the Change Management training programme to have been delivered to all 4,200 employees before the new HR portal goes live.

The Customer Portal requires a customer communications campaign to have been completed at least 30 days before launch, informing customers of the new login experience.

The Change Advisory Board requires impact assessments from all workstreams before scheduling any production change.

The Communications team produces the customer communications campaign material consumed by the Customer Portal pre-launch checklist.

The Training team produces the employee training programme consumed by the HR Integration Service deployment checklist.

Change Management activities begin on 1 February 2026 and run continuously through to programme close on 31 March 2027.

A change readiness survey is scheduled for 1 May 2026 and must be completed before the Integration Phase begins.

Change readiness score must be above 70% for the Integration Phase gate to open.

16. Phase 1 ? Foundation Schedule (February 2026 to June 2026)

Phase 1 Foundation begins on 1 February 2026 and is scheduled to end on 30 June 2026.

Key milestones and their committed dates are as follows.

Core Database cluster provisioning: start 1 February 2026, end 31 March 2026.

Network Infrastructure Upgrade: start 1 March 2026, end 31 May 2026.

Identity Platform deployment: start 1 March 2026, end 1 April 2026.

Data Encryption Module delivery: start 1 February 2026, end 15 March 2026.

Security Audit of Identity Platform: start 15 April 2026, end 30 June 2026.

Privacy Board impact assessment review: start 1 February 2026, end 30 April 2026.

Phase 1 gate review is scheduled for 1 July 2026 and requires all Foundation milestones to be complete and signed off by the Compliance Authority.

Phase 2 Integration begins on 1 May 2026 and must start after Phase 1 Foundation ends.

Phase 1 ends on 30 June 2026.

Phase 2 starts on 1 May 2026.

Phase 2 must start after Phase 1 ends, but Phase 2 is scheduled to start 60 days before Phase 1 ends.

This represents 60 days of negative slack on the phase boundary.

17. Phase 2 ? Integration Schedule (May 2026 to September 2026)

Phase 2 Integration begins on 1 May 2026 and is scheduled to end on 30 September 2026.

Key milestones within Phase 2 are as follows.

API Gateway deployment: start 1 May 2026, end 31 May 2026.

Load Balancer Cluster configuration: start 1 June 2026, end 15 June 2026.

Data Migration (Oracle to PostgreSQL): start 1 April 2026, end 31 July 2026.

User Acceptance Testing: start 1 June 2026, end 31 August 2026.

Training Data Repository population: start 1 May 2026, end 31 July 2026.

Change Readiness Survey: scheduled 1 May 2026.

Integration Testing: start 1 June 2026, end 31 August 2026.

User Acceptance Testing must start after Data Migration ends.

Data Migration ends on 31 July 2026.

User Acceptance Testing starts on 1 June 2026.

User Acceptance Testing must start after Data Migration ends, but it is scheduled to start 60 days before Data Migration ends.

This is 60 days of negative slack.

Integration Testing must start after the API Gateway is deployed.

API Gateway deployment ends 31 May 2026 and Integration Testing starts 1 June 2026; this dependency has zero slack and is on the critical path.

18. Phase 3 ? Validation Schedule (July 2026 to November 2026)

Phase 3 Validation begins on 1 July 2026 and is scheduled to end on 30 November 2026.

Key milestones within Phase 3 are as follows.

Disaster Recovery System provisioning: start 1 July 2026, end 31 August 2026.

Security Certification penetration test: start 15 June 2026, end 15 August 2026.

Security Certification issuance: start 15 August 2026, end 15 November 2026.

Go-Live Rehearsal: start 1 July 2026, end 31 August 2026.

Regulatory Approval Gate review: scheduled 1 October 2026.

Production Deployment go-live: start 1 October 2026, end 31 March 2027.

Production Deployment must start after Security Certification is issued.

Security Certification is not issued until 15 November 2026.

Production Deployment is scheduled to start on 1 October 2026.

Production Deployment must start after Security Certification ends, but is scheduled to start 45 days before Security Certification is complete.

This is 45 days of negative slack.

Go-Live Rehearsal must start after Integration Testing ends.

Integration Testing ends on 31 August 2026.

Go-Live Rehearsal starts on 1 July 2026.

Go-Live Rehearsal must start after Integration Testing ends, but starts 61 days before Integration Testing ends.

This is 61 days of negative slack on this dependency.

19. Programme Risk Register

Risk R-01 (Critical): Privacy Board blocks Identity Platform deployment.

The Privacy Board objection to the Identity Platform architecture, if not resolved by 1 April 2026, will cascade to SSO Module, MFA Gateway, HR Integration Service, Finance Integration Module, Customer Portal, and Audit System - six workstreams blocked.

Probability: High.

Impact: Critical.

Owner: Chief Privacy Officer.

Due date: Resolution required by 15 March 2026.

Risk R-02 (Critical): GDPR Framework blocks Training Data Repository.

If the DPIA is not submitted by 1 April 2026, the AI Platform cannot be operational by 1 August 2026, triggering Machine Learning Pipeline delays and the £3.2 million enterprise customer penalty exposure.

Probability: High.

Impact: Critical.

Owner: Data Protection Officer.

Due date: DPIA submission required by 1 April 2026.

Risk R-03 (High): SLA Compliance blocks Primary Production System migration.

The 4-to-6-hour maintenance window required for cloud migration conflicts with the 99.95% uptime SLA.

Without the blue-green migration budget of £240,000, Cloud Migration and Disaster Recovery are both at risk.

Probability: Medium.

Impact: High.

Owner: Infrastructure Director.

Due date: Budget decision required by 1 March 2026.

Risk R-04 (High): Compliance Authority blocks Data Centre Decommission.

The statutory record-keeping obligation may prevent Data Centre Decommission before 31 March 2027, breaching the budget envelope by an estimated £480,000 in extended lease costs.

Probability: Medium.

Impact: High.

Owner: General Counsel.

Due date: Legal advice required by 1 March 2026.

Risk R-05 (High): Third-Party Integration Framework blocks Legacy API Deprecation.

Vendor clause 7.4 requires 12 months notice; earliest migration is 1 March 2027, 19 months after the planned 30 September 2026 deprecation date.

Probability: High.

Impact: High.

Owner: Vendor Management Director.

Due date: Clause 7.4 renegotiation or waiver required by 1 April 2026.

Risk R-06 (Medium): Security Certification Body blocks Cloud Migration.

Earliest certification issuance is 10 August 2026, making the 1 June 2026 Cloud Migration start date unachievable.

Probability: High.

Impact: Medium.

Owner: Security Director.

Due date: Certification timeline confirmed by 1 February 2026.

Risk R-07 (Medium): Network supplier lead time delay.

100GbE line cards on 14-week order cycle; 15% risk of 4-week delay impacting Cloud Migration, Video Conferencing, Secure Remote Access, and VPN Gateway.

Probability: Low.

Impact: Medium.

Owner: Network Procurement Manager.

Due date: Order confirmation required by 1 January 2026.

20. Resource Allocation and Capacity Plan

The programme requires 87 internal FTE and 34 contractor roles across 22 months.

The Database Administration Team (8 FTE) is fully allocated to Core Database provisioning from 1 February 2026 to 31 March 2026, then to Data Migration from 1 April 2026 to 31 July 2026, then to Integration Testing support from 1 August 2026 to 31 August 2026.

The Database Administration Team has zero contingency capacity during these periods. Any unplanned work arriving from the Analytics Engine or Reporting Module during Data Migration will require explicit reprioritisation by the Nexus Steering Board.

The Security Team (6 FTE) is allocated to Data Encryption Module from 1 February 2026 to 15 March 2026, to Security Audit from 15 April 2026 to 30 June 2026, and to penetration test support from 15 June 2026 to 15 August 2026.

The Security Team requires the Data Encryption Module to be complete before Security Audit begins.

The Data Encryption Module produces the encryption key manifest that Security Audit requires.

The Security Team is therefore self-blocking: if the Data Encryption Module slips beyond 15 April 2026, the Security Team's own Security Audit window is compressed.

The Network Engineering Team (12 FTE) is allocated exclusively to Network Infrastructure Upgrade from 1 March 2026 to 31 May 2026, with no capacity for other work.

The AI/ML Team (10 FTE) begins Training Data Repository setup on 1 May 2026 and transitions to AI Platform deployment from 1 July 2026.

The DevOps Team (9 FTE) provisions the CI/CD pipeline from 1 April 2026 and the Container Orchestration platform from 1 May 2026.

21. Programme Budget Summary

Total approved programme budget: £47,200,000.

Phase 1 Foundation: £12,400,000 (budget release date: 1 February 2026).

Phase 2 Integration: £15,800,000 (budget release date: 1 May 2026).

Phase 3 Validation: £9,600,000 (budget release date: 1 July 2026).

Phase 4 Production Deployment: £9,400,000 (budget release date: 1 October 2026).

Budget release for each phase requires the Nexus Steering Board approval and the Compliance Authority Gate Approval Certificate.

The Nexus Steering Board produces the budget release authorisation consumed by the

Finance Integration Module's cost centre allocation process.

Unresolved risks carry a combined expected value of £7,840,000 against a contingency reserve of £3,200,000, leaving a residual exposure of £4,640,000 that is not currently funded in the approved budget.

The blue-green migration option (Risk R-03) costs £240,000 and is not in the budget.

The extended data centre lease (Risk R-04) costs up to £480,000 and is not in the budget.

The enterprise customer SLA penalties (Risk R-02) are £3.2 million and are not insured.

The vendor API contract renegotiation (Risk R-05) carries legal costs estimated at £120,000.

22. Full Programme Dependency Map

The following relationships are the formal programme dependency register.

SSO Module REQUIRES Identity Platform.

MFA Gateway REQUIRES Identity Platform.

HR Integration Service REQUIRES Identity Platform.

Finance Integration Module REQUIRES Identity Platform.

Customer Portal REQUIRES Identity Platform.

Audit System REQUIRES Identity Platform.

Privacy Board BLOCKS Identity Platform.

Analytics Engine REQUIRES Core Database.

Reporting Module REQUIRES Core Database.

Customer Portal REQUIRES Core Database.

API Gateway REQUIRES Core Database.

Archive System REQUIRES Core Database.

Data Warehouse REQUIRES Core Database.

Cloud Migration REQUIRES Network Infrastructure Upgrade.

Video Conferencing Platform REQUIRES Network Infrastructure Upgrade.

Secure Remote Access REQUIRES Network Infrastructure Upgrade.

VPN Gateway REQUIRES Network Infrastructure Upgrade.

AI Platform REQUIRES Training Data Repository.

Machine Learning Pipeline REQUIRES AI Platform.

BI Reporting Suite REQUIRES Data Warehouse.

Disaster Recovery System REQUIRES Primary Production System.

Container Orchestration REQUIRES DevOps Pipeline.

Service Mesh REQUIRES Container Orchestration.

Load Balancer Cluster REQUIRES API Gateway.

Unified API Layer REQUIRES Legacy API Deprecation.

Cloud Migration REQUIRES Security Certification.

Compliance Authority BLOCKS Data Centre Decommission.

GDPR Compliance Framework BLOCKS Training Data Repository.

SLA Compliance BLOCKS Primary Production System Shutdown.

Third-Party Integration Framework BLOCKS Legacy API Deprecation.

Security Certification Body BLOCKS Cloud Migration.

Cloud Migration REQUIRES Data Centre Decommission.

Production Deployment REQUIRES Security Certification.
Production Deployment REQUIRES Integration Testing.
Integration Testing REQUIRES API Gateway.
Regulatory Hold BLOCKS Production Deployment.
DevOps Pipeline REQUIRES API Gateway.

23. Temporal Dependency and Schedule Sequence Register

The following STARTS_AFTER relationships are recorded in the programme schedule.

Phase 2 Integration STARTS_AFTER Phase 1 Foundation.
Phase 1 Foundation ends on 2026-06-30.
Phase 2 Integration starts on 2026-05-01.
Phase 2 starts before Phase 1 ends: negative slack of 60 days.
User Acceptance Testing STARTS_AFTER Data Migration.
Data Migration ends on 2026-07-31.
User Acceptance Testing starts on 2026-06-01.
UAT starts before Data Migration ends: negative slack of 60 days.
Production Deployment STARTS_AFTER Security Certification.
Security Certification ends on 2026-11-15.
Production Deployment starts on 2026-10-01.
Production Deployment starts before Security Certification ends: negative slack of 45 days.
Go-Live Rehearsal STARTS_AFTER Integration Testing.
Integration Testing ends on 2026-08-31.
Go-Live Rehearsal starts on 2026-07-01.
Go-Live Rehearsal starts before Integration Testing ends: negative slack of 61 days.
Machine Learning Pipeline STARTS_AFTER AI Platform.
AI Platform ends on 2026-08-01.
Machine Learning Pipeline starts on 2026-09-01.
This dependency has positive slack of 31 days.
Disaster Recovery System STARTS_AFTER Primary Production System cloud migration.
Primary Production System migration ends on 2026-06-30.
Disaster Recovery System starts on 2026-07-01.
Positive slack of 1 day.
Network Infrastructure Upgrade must complete before Cloud Migration begins.
Network Infrastructure Upgrade ends 2026-05-31.
Cloud Migration starts 2026-06-01.
Zero slack on this dependency · any network delay directly delays cloud migration.

24. Regulatory and Compliance Framework

The programme is subject to the following regulatory instruments, each imposing mandatory constraints on programme activities.

UK GDPR (2018): governs all processing of personal data.
The GDPR Compliance Framework blocks Training Data Repository population until DPIA sign-off is received.
Financial Services and Markets Act: requires the Audit System to retain a 7-year

authentication log accessible at all times.

The Audit System requires Identity Platform.

Network and Information Systems Regulations (NIS): requires Secure Remote Access and VPN Gateway to use approved cryptographic standards.

Both require Network Infrastructure Upgrade.

ISO 27001 certification: the Security Certification Body requires evidence of ISO 27001 compliance before issuing Security Certification for Cloud Migration.

The Compliance Authority produces the Gate Approval Certificate at four programme checkpoints: Foundation Gate (1 July 2026), Integration Gate (1 October 2026), Validation Gate (1 December 2026), and Production Gate (1 January 2027).

Each Gate Approval Certificate requires a completed assurance package from the Security Team and a Programme Authority Notice from the Nexus Steering Board.

Regulatory Hold is automatically invoked if any gate is missed or any open compliance finding is not remediated within 30 days of being raised.

Regulatory Hold blocks Production Deployment, blocks API Gateway deployment, and blocks Legacy System Shutdown simultaneously.

25. Target State Technical Architecture

The target architecture is a fully cloud-native microservices platform hosted on the contracted hyperscaler across three regions: UK South (primary), UK West (secondary), and EU West (compliance-mandated data residency).

The Identity Platform hosts the OAuth 2.0 authorisation server, the OIDC provider, and the SAML 2.0 bridge required for legacy application federation.

The Core Database cluster runs PostgreSQL 16 with Citus horizontal sharding, a read replica set for the Analytics Engine, and a change-data-capture stream feeding the Data Warehouse via Apache Kafka.

The Network Infrastructure Upgrade provisions BGP route reflectors, EVPN-VXLAN overlays, and 100GbE spine switches with SR-IOV NIC passthrough for GPU workloads in the AI cluster.

The AI Platform runs on a dedicated GPU node pool with A100 instances, managed by the Container Orchestration platform and exposed via the Unified API Layer.

The Service Mesh provides mutual TLS, circuit breaking, and distributed tracing across all microservices, feeding telemetry to the BI Reporting Suite.

The DevOps Pipeline integrates with the Container Orchestration platform to provide GitOps-driven deployment, policy-as-code enforcement, and automated security scanning for every container image before it reaches the production registry.

26. Vendor and Contract Management

The programme engages 14 external vendors under a mix of fixed-price and time-and-materials contracts.

The Third-Party Integration Framework manages the API endpoint commitments made to each vendor under their respective master service agreements.

Vendor contract clause 7.4 requires 12 months' written notice before any API endpoint URL change.

This clause directly conflicts with the Legacy API Deprecation milestone.

The Unified API Layer requires Legacy API Deprecation; the Third-Party Integration Framework blocks Legacy API Deprecation.

Vendor renegotiation is in progress for 3 of the 14 vendors.

The remaining 11 vendors have not been approached.

The Vendor Management Director has estimated that renegotiation will take 6 months per vendor.

This is incompatible with the 30 September 2026 Legacy API Deprecation date.

The cloud provider contract was signed on 1 December 2025 for a 36-month term commencing 1 February 2026, at a total committed spend of £18,400,000.

The GPU node pool for the AI Platform carries a minimum commitment of £2,400,000 per year regardless of utilisation, meaning the AI Platform cost is incurred from 1 February 2026 even if the platform is not operational by its scheduled date.

27. Data Governance and Classification

All data assets processed by the Nexus Programme are classified under the organisation's four-tier data classification scheme: Public, Internal, Confidential, and Restricted.

The Training Data Repository contains customer transaction records classified as Restricted under the data classification scheme.

The GDPR Compliance Framework requires that Restricted data may not be used for AI model training without explicit data subject consent or an alternative lawful basis.

The Data Protection Officer has not yet identified a lawful basis for processing the Training Data Repository contents for model training.

The Data Warehouse contains Internal and Confidential data fed from Core Database.

The BI Reporting Suite requires the Data Warehouse and must implement role-based access control enforced by the Identity Platform.

The Archive System retains Confidential and Restricted data for the mandatory retention periods defined by the data classification policy: 7 years for financial records, 3 years for operational records, and 1 year for system event logs.

The Archive System requires Core Database for its retention metadata catalogue.

The Audit System requires Identity Platform to associate all data access events with authenticated user identities for the forensic audit trail required by regulators.

28. Security Architecture and Controls

The programme's security architecture is governed by the Security Framework produced by the Security Team and approved by the Security Certification Body.

The Data Encryption Module produces the encryption key manifest consumed by the Security Audit and required by the Identity Platform's token signing infrastructure.

Security Audit requires Data Encryption Module to be complete and the key manifest to be available before the audit scope can be confirmed.

The Security Certification Body requires Security Audit sign-off as a prerequisite for issuing Security Certification for Cloud Migration.

Zero-trust network access is enforced by Secure Remote Access, which requires Network Infrastructure Upgrade for its NAC policy enforcement points.

All inter-service communication is encrypted at the transport layer by the Service

Mesh, which requires Container Orchestration to be operational.
The MFA Gateway enforces step-up authentication for all Restricted data access, requiring Identity Platform to be healthy.
Penetration testing is scheduled to begin 15 June 2026 against the full cloud environment, after Network Infrastructure Upgrade completes on 31 May 2026.
Security Certification is expected by 15 November 2026, assuming a 3-month remediation cycle following the penetration test.

29. Phase Gate Acceptance Criteria

Foundation Gate (1 July 2026) acceptance criteria: Core Database cluster provisioned and health-checked.
Network Infrastructure Upgrade complete and signed off by Network Engineering Team.
Identity Platform deployed (subject to Privacy Board resolution).
Data Encryption Module delivered and key manifest generated.
Security Audit completed and report signed off by Security Team.
Privacy Board impact assessment complete and decision recorded.
Compliance Authority Gate Approval Certificate issued.
Integration Gate (1 October 2026) acceptance criteria: API Gateway deployed and traffic routing validated.
Load Balancer Cluster configured and tested under load.
Data Migration complete and data integrity verified.
User Acceptance Testing complete with defect count below threshold.
Integration Testing complete and system validation report signed off.
Training Data Repository populated (subject to GDPR sign-off).
Change Readiness Survey score above 70%.
Validation Gate (1 December 2026) acceptance criteria: Disaster Recovery System operational and DR test passed.
Security Certification issued by Security Certification Body.
Go-Live Rehearsal completed successfully.
Regulatory Approval Gate review passed.
Production Gate (1 January 2027) acceptance criteria: All Foundation, Integration, and Validation gate criteria confirmed still met.
Production Deployment plan approved by Change Advisory Board.
Compliance Authority final Gate Approval Certificate issued.

30. Programme Closure and Benefits Realisation

The programme will be formally closed when all of the following conditions are met:
All phase gate criteria are confirmed as met and documented in the programme closure report.
Production Deployment is complete and the system has been in stable production operation for a minimum of 30 days without a severity-1 incident.
All legacy systems have been decommissioned or formally handed over to sustain teams.
Data Centre Decommission is complete or an approved deferral plan is in place.
All vendor transitions under the Third-Party Integration Framework are complete or contractual waivers for Legacy API Deprecation are in place.

Benefits realisation review is scheduled for 1 April 2027 and will assess whether the committed benefits of £14,600,000 annualised cost reduction have been achieved. The Identity Platform unified authentication is expected to deliver £2,800,000 per year by eliminating password reset incidents and identity administration overhead. The Core Database consolidation is expected to deliver £4,200,000 per year by retiring Oracle licensing and the 12 point-to-point data feed maintenance contracts. The Network Infrastructure Upgrade and Cloud Migration together are expected to deliver £7,600,000 per year by retiring the data centre lease and reducing network operations staffing.

Programme closure is targeted for 31 March 2027.

This charter was approved by the Nexus Steering Board on 1 February 2026.

Document owner: Programme Director.

Review cycle: monthly.